SECTION A

1. (a) A beam ABC is subjected to a clockwise moment M at point C as shown in Figure 1(a).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tension** side. **(5 marks)**

   (b) A beam ABCD is subjected to a uniformly distributed load q along CD. The beam segment AB and CD are connected by a hinge joint at point B as shown in Figure 1(b).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tension** side. **(5 marks)**

2. A concrete beam AB is suspended to the ceiling by two steel wire strings AC and BD as shown in Figure 2(a). The cross-section of steel beam is rectangular with breadth b = 0.20 m and height h = 0.30 m. The length of concrete beam is L = 5 m and density ρ= 24 kN/m^3. The diameter of the steel wire d is 0.01 m and the length l is 0.1 m. The Young's modulus of steel wire and concrete is E_1 = 200 GPa and E_2 = 30 GPa respectively. (Note: Moment of inertia for rectangular cross-section is $I=(b\times h^3)/12$)

   (a) Draw the bending moment diagram (BMD) for concrete beam AB under gravity load (uniformly distributed load of the beam due to gravity). Note that bending moment should be drawn on the **tension** side. **(3 marks)**

   (b) Determine the elongation of strings AC and BD (Note: gravity load of steel wire can be ignored). **(3 marks)**

   (c) Sketch the deformation of beam AB and determine the maximum deflection at center point E subjected to gravity load. **(5 marks)**

   (d) As shown in Figure 2(b), if we add point load P on center of beam AB and the allowable tensile strength of concrete beam $[\sigma]$ is 3.0 MPa. Determine the critical point load P_{max} and point out the position where crack occur first. (Hint: crack appears due to maximum tension stress) **(6 marks)**

   (e) If we add one more steel wire string EF in the middle of the beam as shown in Figure 2(c). Sketch the deformation of beam AEB due to gravity load. Assume that all the steel wire strings have infinite stiffness (i.e., $E\rightarrow\infty$). **(3 marks)**

3. The point load P is placed at the end of the steel beam ADC with T-section as shown in Figure 3(a). A steel brace BD is applied to bear the axial load only with the circular cross section illustrated in Figure 3(b). The steel brace BD is connected to steel beam ADC by hinge joint at D. The Young's modulus of steel beam and brace is 200 GPa. (Note: Moment of inertia for circular cross-section is $I=(\pi \times d^4)/64$)

(a) Considering the Euler's buckling of steel brace BD, find the critical point load P_{max} at point C. **(5 marks)**

(b) For steel beam ADC, draw axial force diagram (AFD), shear force diagram (SFD) and bending moment diagram (BMD). **(6 marks)**

(c) When P = 1 kN, determine the maximum bending stress for beam ADC (both tension and compression stress) and point out their locations (Hint: Axial deformations can be ignored). **(4 marks)**

(d) When P = 1 kN, determine contraction of steel brace BD. **(2 marks)**

(e) If steel beam ADC has infinite bending stiffness $EI_{ADC} \to \infty$, find the deflection at point C. **(3 marks)**

Hint: According to Euler's formula, the critical load for a slender column with span L is $P_{cr} = \frac{\pi^2 EI}{(\mu L)^2}$, where μ is length coefficient. μ=1.0 for both pinned ends, μ=2.0 for one fixed end and one free end, μ=0.7 for one fixed end and one pinned end, and μ=0.5 for both fixed ends.

FIGURES

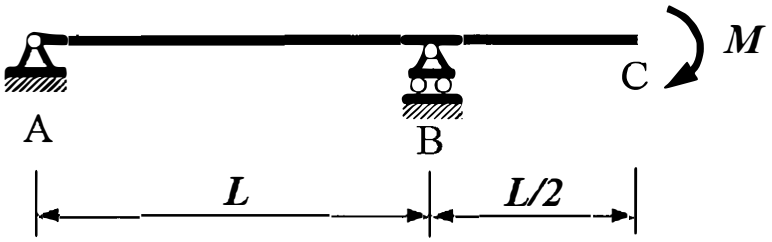


Figure 1(a) Beam

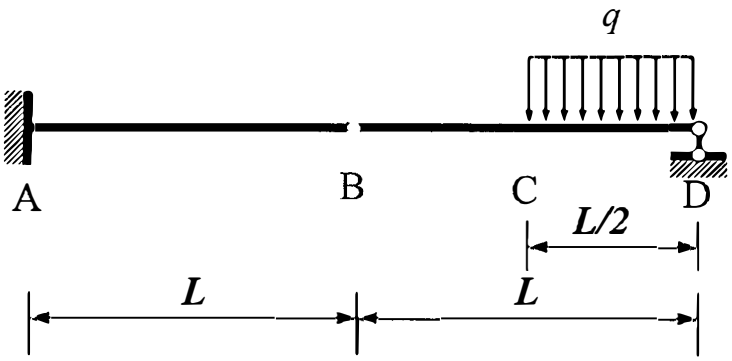


Figure 1(b) Beam

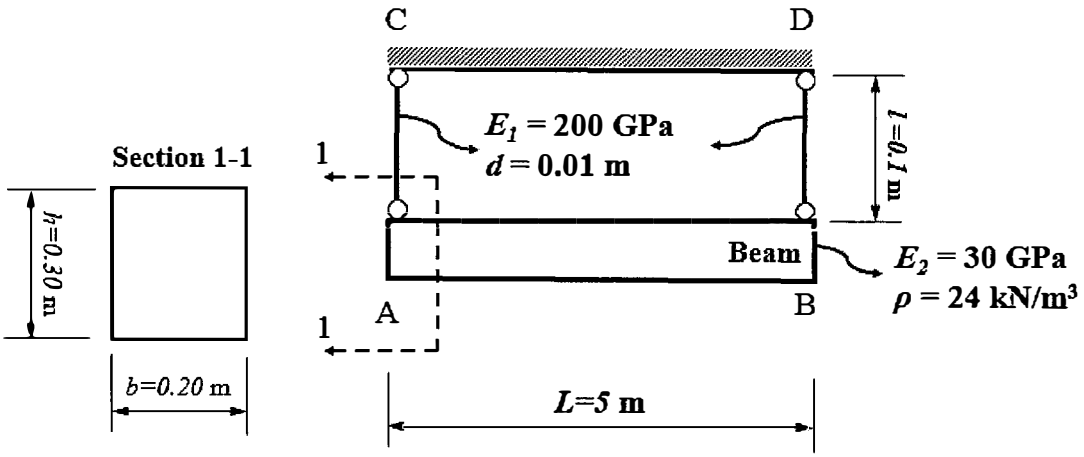


Figure 2(a) Suspended beam

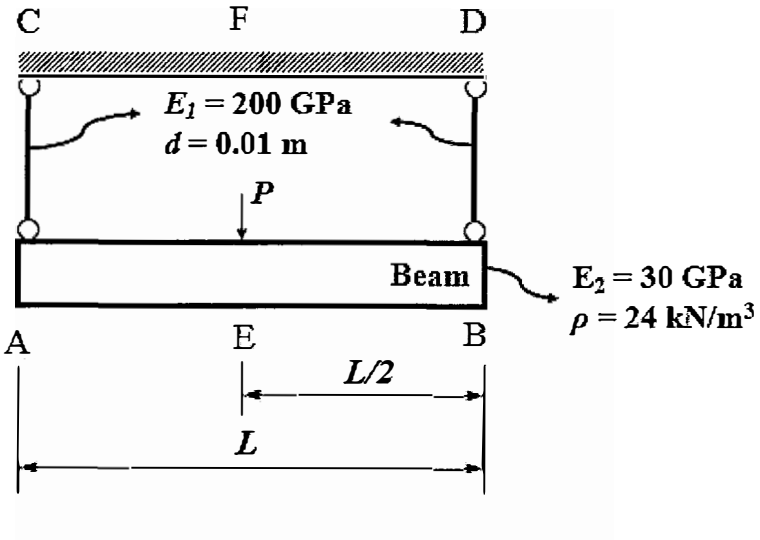


Figure 2(b) Suspended beam with point load *P*

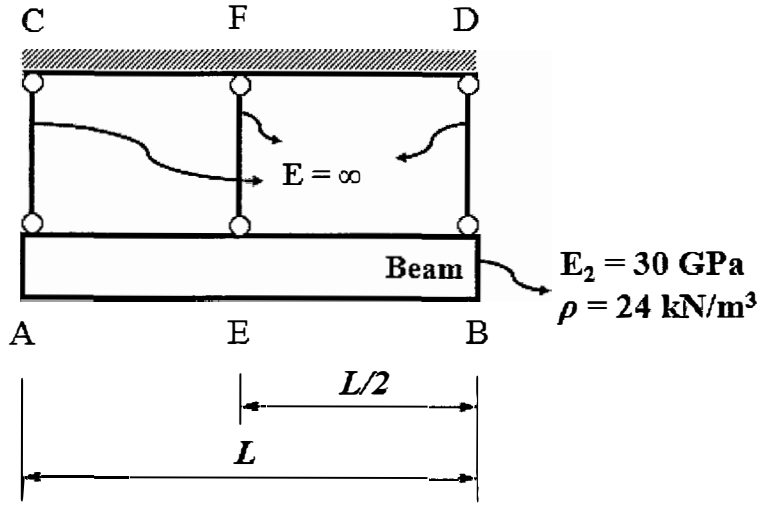


Figure 2(c) Suspended beam with three steel wires